Distribution (With Mod S.O.O. 6247) – General Data

1. General

Fuel distribution provides fuel transfer within the fuselage tanks, and pressure fuel distribution from the fuselage and wing tanks to the engines. The right engine obtains its pressure fuel supply from the forward fuselage and right wing tanks, the left engine from the aft fuselage and left wing tanks. A crossfeed valve when selected allows both engines to be fed from one fuselage tank and its associated wing tank.

2. Description

The fuel distribution system consists of fuel booster pumps, wing fuel pressure pumps, wing tank fuel filter, a crossfeed valve, emergency shut-off valves, fuel transfer valves, fuel filters, control switches, fuel ejectors, drain valves, level control valves, and check valves. Aircraft have arrows stenciled adjacent to each check valve in the fuel manifold, indicating direction of fuel flow. This ensures correct installation of the check valves when the arrow marked on each check valve coincides with its related stenciled arrow. The pressure fuel supply line to each engine is routed from a fuselage booster pump to No. 6 cell, through the cell, up the rear of a wing strut, along the wing into the engine nacelle where it joins the wing tank pressure fuel supply line. From this junction, at the inlet side of the fuel filter, the fuel is routed through the fuel filter element, a fuel emergency shut-off valve, and through the firewall to the engine-driven fuel pump. The electrical power to the system is supplied from the 28 Volt DC bus.

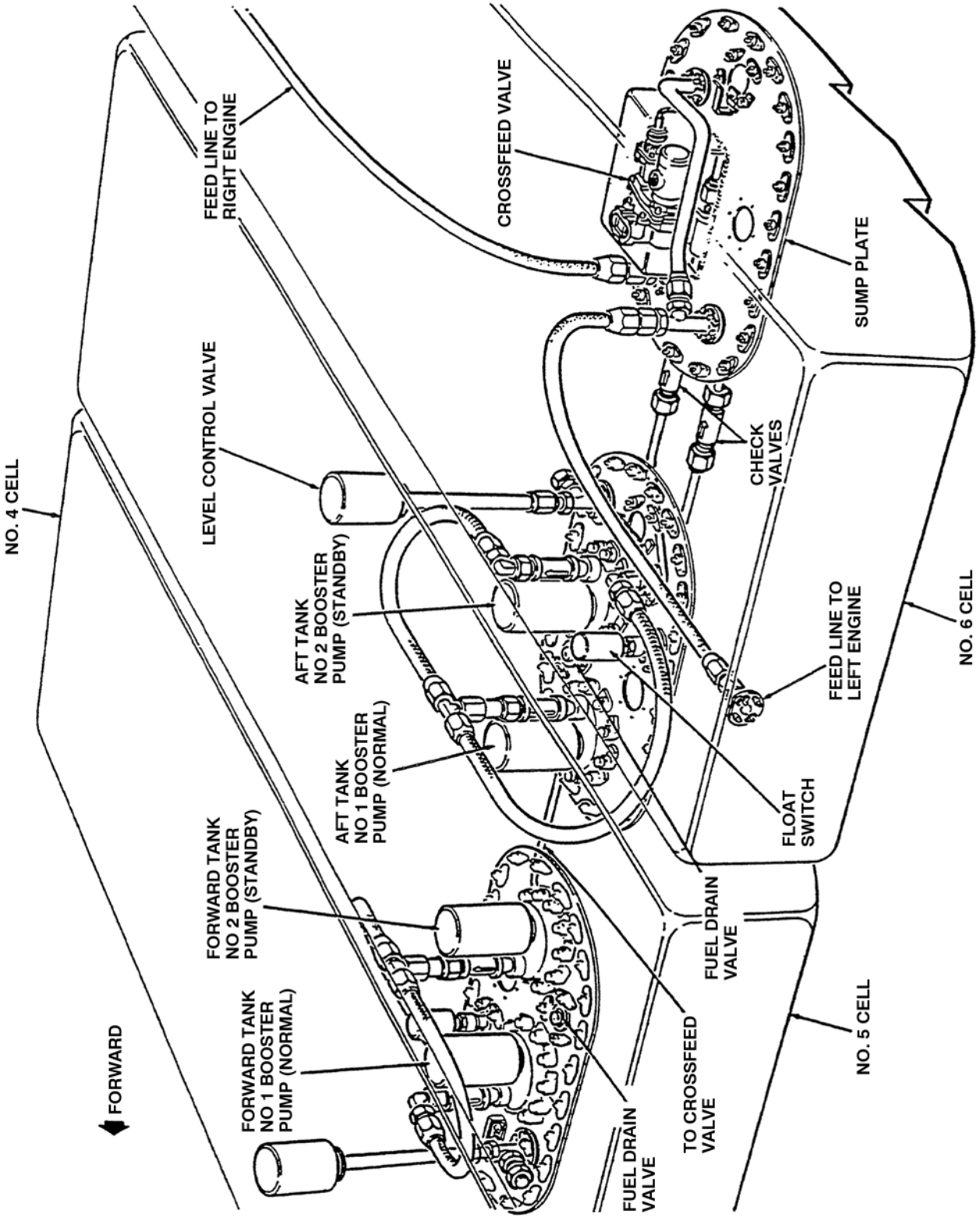
A. Fuel Booster Pumps

Four fuel booster pumps are installed, two in No. 4 cell of the forward tank and two in No. 5 cell of the aft tank. On aircraft with Mod 6/1795 installed, there are two submerged constant speed, centrifugal, electric fuel boost pumps and are located in each fuselage tank collector cell. The pumps deliver low-pressure fuel. They are powered from the left and right DC buses and are protected by circuit breakers labelled BOOST PUMP AFT 1, BOOST PUMP AFT 2, BOOST PUMP FWD 1 and BOOST PUMP FWD 2 on the main circuit breaker panel. Pre Mod 6/2267 pumps contain Captor Corp., part number A–3210 EMI filters installed in the circuit. The filters are mounted to the pump base. Post Mod 6/2267 pumps incorporate Lone Star, part number 122253–10A EMI filters installed in the circuit. The filters are mounted on a sheet metal plate installed on the RH Cabin sidewall at approximately station 135, below the window. On aircraft with Mod 6/2299 installed, improved fuel pumps have a new EMI/RFI electrical noise filter and ground circuit to improve corrosion tolerance of the pump and nearby components. Mod 6/2299 fuel pumps have two electrical fly leads that are connected to the aircraft wiring.

The fuel tank selector is located on the center instrument panel and marked FUEL SELECTOR. It is a rotary knob with three marked positions – BOTH ON FWD, NORMAL and BOTH ON AFT. In the NORMAL position the forward and aft tanks supply the right and left engines, respectively, the crossfeed valve is closed, and the boost pump(s) in each tank is energized . In the BOTH ON FWD or AFT position, the crossfeed valve is open, the boost pumps of the non selected tank are automatically deenergized, and both boost pumps in the selected tank are energized (the boost pump in the selected tank are energized (the boost pump switch selection is overridden). Boost pump switches located on the fuel panel are labelled BOOST PUMP 1 with ON, OFF and TEST positions marked. With power ON, circuit breakers set, and the FUEL SELECTOR positioned to NORMAL, the No. 1 boost pump in each collector tank is energized when ON position is selected; the No. 2 boost pumps remain inoperative. Control of the No. 2 boost pump in each tank is by means of an automatic electrical changeover sequence, which is initiated by the failure of the No. 1 boost pump. Refer to Figure 1 and Table 1 — Fuel Booster Pumps – Modification State.

Table 1 Fuel Booster Pumps – Modification State

MODIFICATION NO.	BOOSTER PUMP POSITION		TYPE OF PUMP	EMI FILTER
6/1795	No. 1	No. 2	Airborne (1D2–17)	A–3210 Captor Corp
6/2267	No. 1	No. 2	Airborne (1D2–20)	122253–10A Lone Star
6/2299	No. 1	No. 2	Airborne (1D2–21)	Embedded Filter (Captor Corp)



Booster Pump and Crossfeed Valve Installation
Figure 1

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B. Wing Fuel Pressure Pump

A fuel pressure pump is mounted outside each wing tank on the wing outboard nose rib. The pumps are electric motor-driven and controlled by switches on the fuel control panel. The left wing tank pump is operated when the left switch is selected to left engine; similarly the right switch operates the right wing tank pump. The wing fuel pressure pumps when operating will overcome fuselage booster pump fuel pressure and empty the wing tanks before the fuselage tanks. On aircraft incorporating Mod 6/1374 a radio noise filter is installed in each pump circuit.

C. Crossfeed Valve

The crossfeed valve is an electric motor-operated gate valve which is controlled by the FUEL SELECTOR switch on the fuel control panel in the flight compartment. The valve is mounted in the fuel line on the floor of No. 6 fuel cell. To replace a crossfeed valve it will be necessary to first drain both fuselage fuel tanks.

D. Fuel Emergency Shut-off Valves

There are two fuel emergency shut-off valves provided to shut off the fuel supply in case of engine malfunction. Each valve is an electric motor-operated gate valve mounted in the nacelle behind the firewall. A switch is provided on the emergency panel for each valve and is marked FUEL OFF and NORMAL. To replace an emergency shut-off valve, drain fuel from filter (refer to 28-20-31, Fuel Filter (Mod 6/2006) – Maintenance Practices), then remove valve.

E. Fuel Filter

A fuel filter is provided for each engine and is mounted behind the engine firewall to the rear of the fuel emergency shut-off valve. The fuel filter is an assembly consisting of a housing, filter element and bypass valve. Should the filter element become clogged. The fuel forces the bypass valve to open and the unfiltered fuel passes directly through the center of the filter. A drain valve is installed in the filter cover.

F. Standby Booster Pump Emergency Switches

Two lever-lock type standby booster pump emergency switches, marked, BOOST PUMP 2 are located on the fuel control panel the emergency switches override the automatic change-over system when selected to EMERGENCY to operate the corresponding No. 2 booster pump in the aft and/or forward fuselage tank. The switches are locked in the off position and must be pulled before a selection can be made. Refer to Figure 2 for details.

G. Wing Fuel Tank Switches

Two lever-lock type wing fuel tank switches, marked LEFT WING TANK and RIGHT WING TANK are located on the FUEL SELECTOR panel. There are three selections available on each switch, REFUEL, OFF and ENGINE. The switch lock is in operation in the OFF position and must be pulled before either of the two other selections can be made. When REFUEL is selected, the refuel shut-off valve is opened and allows fuel, from the fuselage booster pumps (if selected), to flow through the level control valve into the wing tank. The ENGINE selection operates the applicable wing tank fuel pressure pump and opens a fuel transfer valve.

H. Fuel Ejectors

The fuel ejectors are designed on the venture principle so that no moving parts are required. Fuel under pressure from the main feed system passing through the ejector draws additional fuel from the tank and moves the total fuel through the transfer line into the main cell. There are two ejectors; one is mounted on the sump plate of No. 1 cell and the other on the sump plate of No. 8 cell.

I. Drain Valves

There are six water drain valves incorporated in the fuel system to provide a check for water accumulation. One is installed on each fuel filter, one in each fuel manifold, and one in each wing tank. Those on the fuel

filters are located in the engine nacelles, the fuel manifold water drains are located in the trough on the underside of the fuselage, and the wing tank water drains protrude from the underside of each wing tank.

J. Fuel Level Control Valves

There are four level control valves incorporated in the fuel system, one in each fuel tank. The fuselage tanks level control valves are mounted on the sump plates of cells No. 4 and 5. Each valve controls the level of fuel in the cell or tank by closing off or admitting fuel from the fuselage transfer system or wing refueling shut-off valve line.

K. Fuel Transfer Valves

A fuel transfer valve is fitted in each wing tank pressure fuel line on the outboard side of the tank. The valve is a solenoid-operated shut-off type valve and is opened when the wing fuel pressure pump is operated by an applicable L or R ENGINE selection on the FUEL SELECTOR wing tank fuel switches.

L. Refuel Shut-off Valve

A refuel shut-off valve is fitted to a tee connection in each wing tank pressure fuel line on the rear outboard side of each tank. When operated, the valve opens the pressure fuel line from the fuselage booster pumps to the level control valve inside the wing fuel tank. This allows booster pump pressure fuel from the fuselage tanks to refuel the wing tanks. A refuel shut-off valve is opened by selecting REFUEL on the wing tank fuel switch.

M. Motive Line Check Valves

There are two motive line check valves fitted in the pressure lines to the ejectors, one beneath No. 1 cell and one beneath No. 7 cell. Each valve housing contains a fuel filter which can be removed for cleaning when the valve is disassembled.

N. Thermal Relief Check Valve

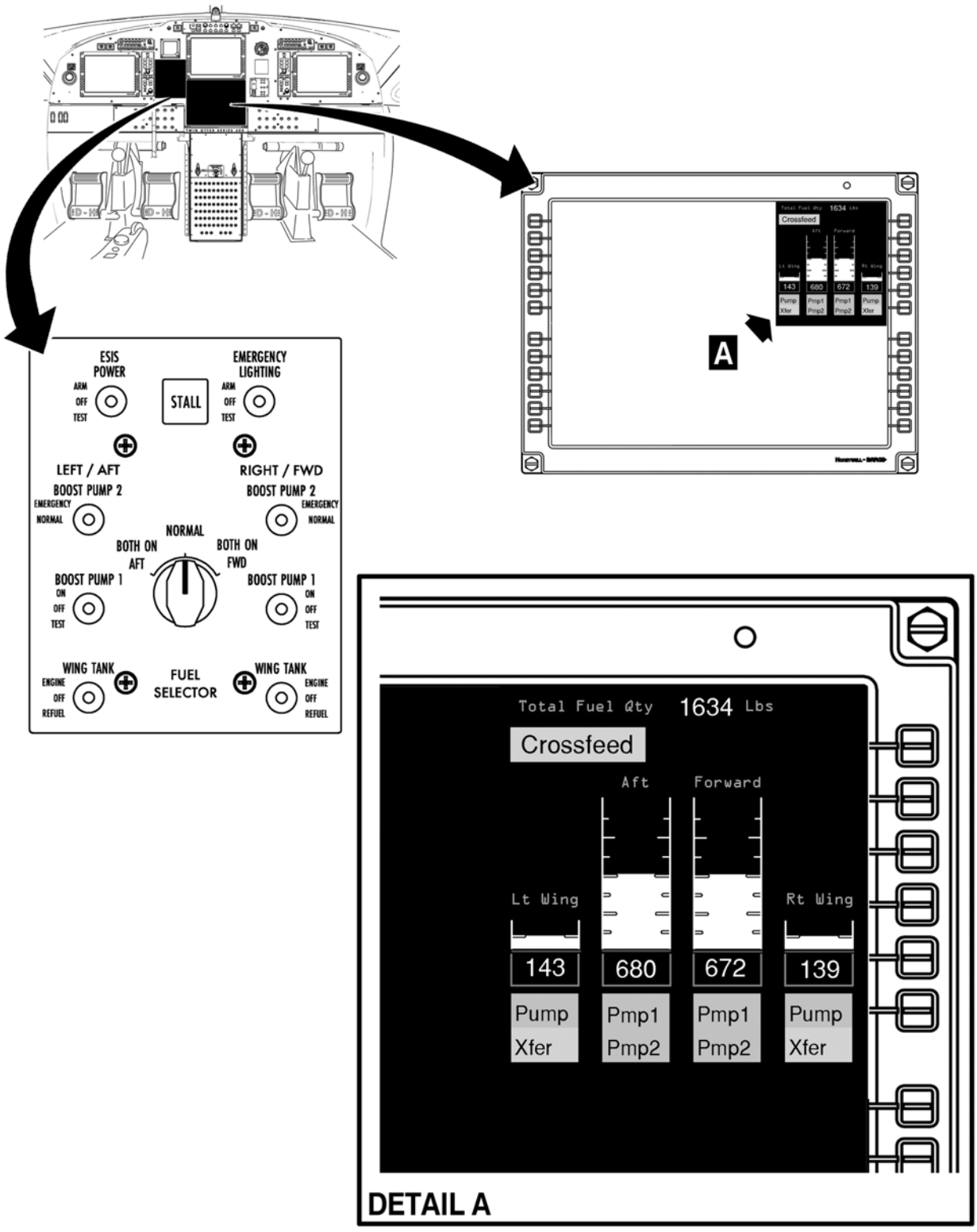
Two thermal relief check valves, one for each tank pressure line, are located below the forward edge of No. 6 fuel cell sump plate. The valve consists of a flap type check valve with an internal bleed provided to prevent thermal pressure buildup in the engine fuel supply line.

O. Fuel Controls

Fuel controls consist of two main booster pump switches (BOOST PUMP 1), two emergency boost pump switches (BOOST PUMP 2), and rotary FUEL SELECTOR switch. The components are all located on the fuel control panel. Fuel controls for the wing tanks consist of two fuel pressure pump switches. The switches are located on the fuel control panel. Indication is displayed on lower 1/6th screen of the fuel window on the MFD. Refer to Figure 2 for details.

P. Wing Tank Filter (Mod 6/1398)

An in-line filter is installed between the wing tank and the wing fuel pressure pump, the filter housing contains a pleated metal filter element which can be removed for cleaning.



Fuel Controls
Figure 2

3. Operation

With the 28 Volt DC bus energized, circuit breakers engaged, FUEL SELECTOR switch at NORMAL, fuel booster pump switches at ON, the wing fuel tank switches at ENGINE and engines running, the system is in the normal operating mode. The wing fuel pressure pumps will supply fuel to the appropriate engine until the wing tanks empty, at which time the No. 1 fuel booster pumps will supply the engines from the fuselage tanks. The wing fuel tank switches should be set to OFF when the wing tanks empty. As the fuel is used from collector cells 4 and 5, fuel is transferred from cells 1 and 8 by booster pump operated ejectors to maintain a maximum fuel level in the collector cells. The fuel level control valves in the collector cells prevent overfilling. As the fuel is transferred from cells 1 and 8, replacement fuel is transferred by gravity from cells 2 and 3 into 1, and from cells 6 and 7 into 8.

When crossfeeding is selected, by moving the FUEL SELECTOR switch to BOTH ON AFT, both engines are supplied with fuel from the aft fuselage and right wing tanks. When BOTH ON FWD is selected, both engines are supplied from the forward fuselage and left wing tanks. If crossfeeding, the pump on the wing tank not in use should be switched off.